Functional Symmetry, Tangency, and Sharp Inequalities

N-20221105

A functional equation can be inconsistent before one attempts to solve it. A polynomial can change the number of roots as a parameter moves. And an inequality may become sharp only when its constraint is used. These simple problems reward structural calculations that will later reappear as group actions, fold singularities, and convex geometry.[1]

§1.1 A translation and a reflection need to be compatible

Consider the drift equation

$$f(x+4) = f(x) + 2.$$

Subtracting the linear function $x/2$ makes the behavior periodic:

$$h(x) = f(x) - \frac{x}{2}, \qquad h(x+4) = h(x).$$

Thus every solution consists of a linear drift plus an arbitrary 4-periodic function.

What happens if we also demand the reflection symmetry

$$f(2-x) = f(x)?$$

There are no solutions. The reflection condition at $x-4$ gives

$$f(6-x) = f(x-4) = f(x) - 2,$$

while translation applied to $2-x$ gives

$$f(6-x) = f(2-x) + 2 = f(x) + 2.$$

The same value cannot equal both $f(x)-2$ and $f(x)+2$.

The structural reason is a relation in the infinite dihedral group. The translations $T(x) = x+4$ and reflection $R(x) = 2-x$ satisfy

$$RTR = T^{-1}.$$

A reflection reverses the translation, so their corresponding transformations of function values must also be compatible. Shifting the output by $+2$ is not compatible with leaving the output unchanged under reflection.

A point-reflection condition *is* compatible:

$$f(x+4) = f(x) + 2, \qquad f(2-x) = 2 - f(x).$$

Its complete solution family is

$$\boxed{f(x) = \frac{x+1}{2} + h(x),} \qquad h(x+4) = h(x), \quad h(2-x) = -h(x).$$

For instance $h(x) = A\sin(\pi(x-1)/2)$ works for every real A. The computation has converted a functional equation into an invariance and anti-invariance problem under two generators.

[1] For the classical inequality methods see Hardy, Littlewood and Pólya, *Inequalities*; the parameter analysis below uses only standard calculus and the cubic discriminant.

§1.2 A logarithm comparison is also an analytic expansion

For $a > 0$, $a \neq 1$, and $-1 < x < 1$, consider

$$F_a(x) = \log_a(1+x) - \log_a(1-x) = \frac{\log((1+x)/(1-x))}{\log a}.$$

The numerator has the same sign as x, so the sign of $F_a(x)$ changes when a crosses 1. In particular, inequalities reverse when the base is between zero and one.

The numerator has a useful expansion. Since

$$\frac{d}{dx}\log\frac{1+x}{1-x} = \frac{2}{1-x^2},$$

integrate the geometric series to obtain, for $|x| < 1$,

$$\boxed{\log\frac{1+x}{1-x} = 2\sum_{n=0}^{\infty}\frac{x^{2n+1}}{2n+1}.}$$

For $0 < x < 1$, the truncation through $n = m$ is a strict lower bound, and the remainder satisfies

$$0 < R_m(x) \leq \frac{2x^{2m+3}}{(2m+3)(1-x^2)}.$$

Thus a simple sign problem naturally yields the Taylor series of $2\operatorname{artanh}(x)$ and a rigorous approximation error.

§1.3 Two roots merge at a tangent level

Consider the parameter-dependent cubic

$$F_p(x) = x^3 - 3px + 2.$$

A tangent zero satisfies both $F_p(x) = 0$ and $F_p'(x) = 0$. The second equation gives $p = x^2$, and substitution into the first gives

$$-2x^3 + 2 = 0.$$

Hence

$$\boxed{x = 1, \qquad p = 1, \qquad F_1(x) = (x-1)^2(x+2).}$$

The root at 1 is double; the root at -2 is simple.

The discriminant of $x^3 + bx + c$ is $-4b^3 - 27c^2$. For this cubic,

$$\Delta(p) = 108(p^3 - 1).$$

The equation has one distinct real root if $p < 1$, two distinct roots when $p = 1$, and three distinct real roots if $p > 1$.

Write $x = 1 + y$ and $p = 1 + \delta$. A direct expansion gives

$$F_{1+\delta}(1+y) = 3y^2 + y^3 - 3\delta(1+y).$$

When $\delta > 0$ is small, the two roots that meet at 1 obey

$$y = \pm\sqrt{\delta} + O(\delta).$$

A polynomial exercise has exposed the local square-root geometry of a fold: solutions appear in pairs across a critical parameter.

§1.4 The product constraint produces a sharp inequality

For $a, b, c > 0$ with $abc = 1$, prove

$$\boxed{\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{3}{2}.}$$

Cauchy–Schwarz in Engel form yields

$$\sum_{\text{cyc}} \frac{a^2}{b+c} \geq \frac{(a+b+c)^2}{2(a+b+c)} = \frac{a+b+c}{2}.$$

AM–GM and $abc = 1$ give $a + b + c \geq 3$, so the desired bound follows. Equality in both steps requires $a = b = c = 1$; the constant is sharp.

There is also a useful local geometric interpretation of the constraint. Put

$$a = e^u, \qquad b = e^v, \qquad c = e^{-u-v}.$$

The condition $abc = 1$ is now automatic. By convexity of the exponential,

$$e^u + e^v + e^{-u-v} \geq 3.$$

A second-order expansion around the equality point gives

$$a + b + c = 3 + \frac{1}{2}\big(u^2 + v^2 + (u+v)^2\big) + O\big((|u| + |v|)^3\big).$$

The positive quadratic correction quantifies how the sum grows away from the equal-variable configuration. In logarithmic coordinates, the product constraint has become a linear plane and AM–GM is simply a Jensen inequality on that plane.

§1.5 Quadratics give the exact local model

For comparison, completing the square in a one-variable problem yields

$$x^2 + x + \frac{5}{3} = \left(x + \frac{1}{2}\right)^2 + \frac{17}{12}.$$

The excess above the minimum $17/12$ is exactly a square. At the constrained exponential minimum the excess is only approximately a positive quadratic form, but it plays the same local role.

The method across these problems is to identify what cannot change. A functional equation respects the relations of its symmetry group, a critical root is marked by the simultaneous vanishing of a function and its derivative, and the equality case of an inequality identifies the point about which a convexity argument can be sharpened. Each is a useful route from elementary algebra to structural mathematics.